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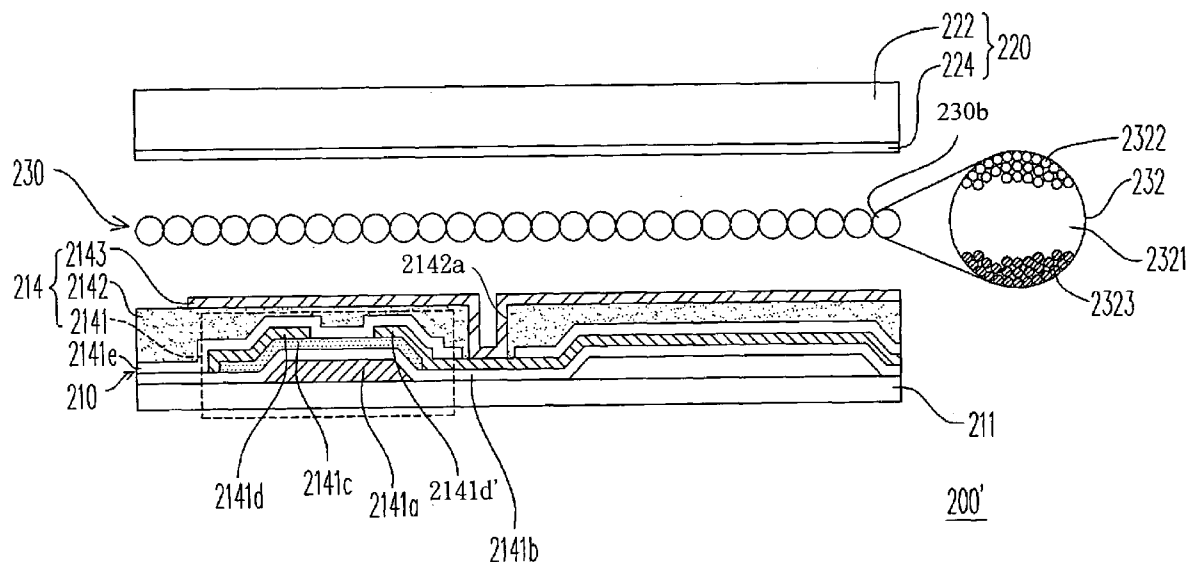
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(57) **ABSTRACT**

An E-ink display panel is provided. In the E-ink display panel, a dielectric layer having a thickness of 1~4 μm is disposed on bottom-gate thin film transistors to prevent the pixel electrodes from being interfered by noise generated by the bottom-gate thin film transistors, the scan lines and the data lines. Therefore, the display quality of the E-ink display panel is improved.

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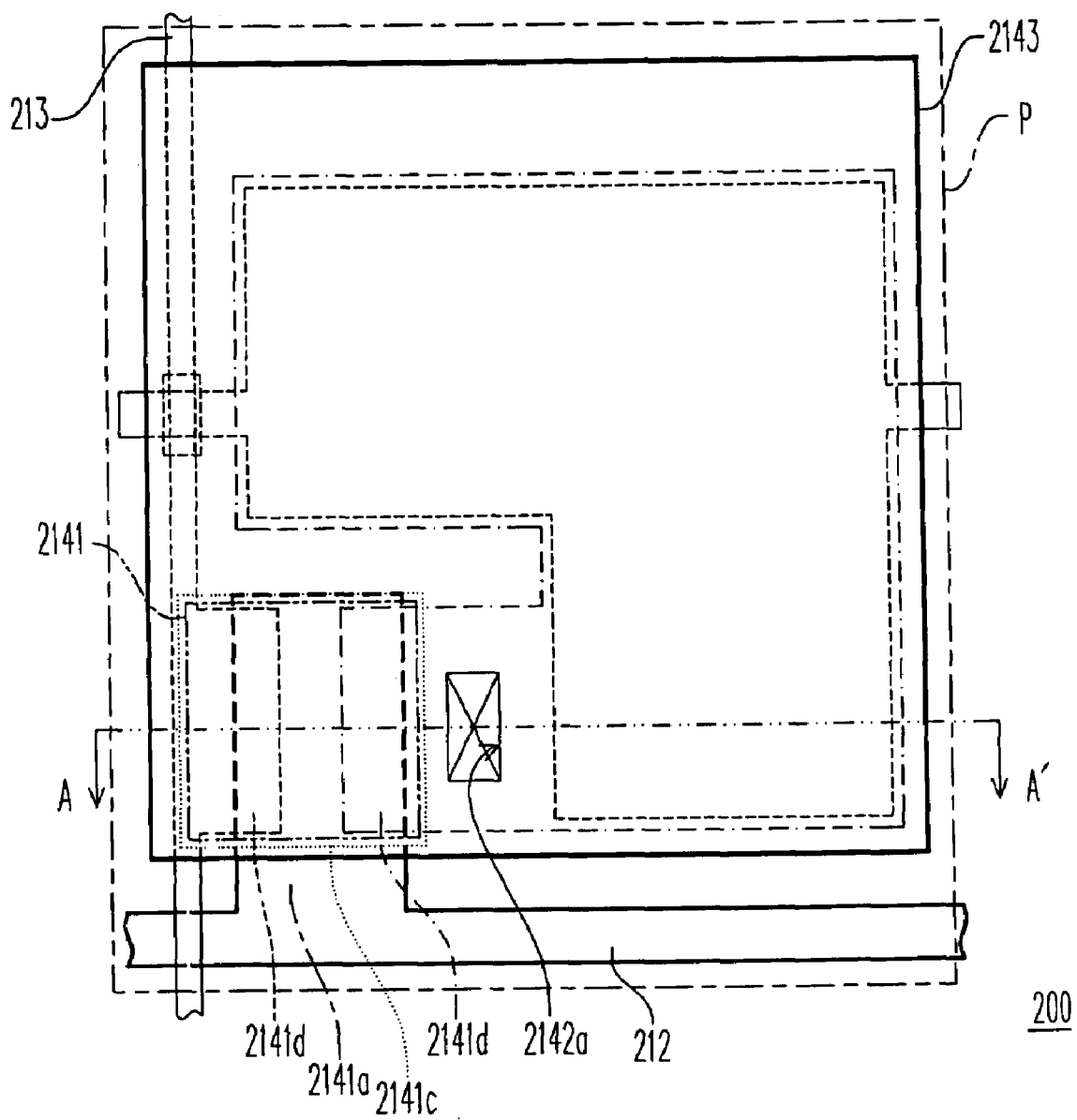


Fig. 1A

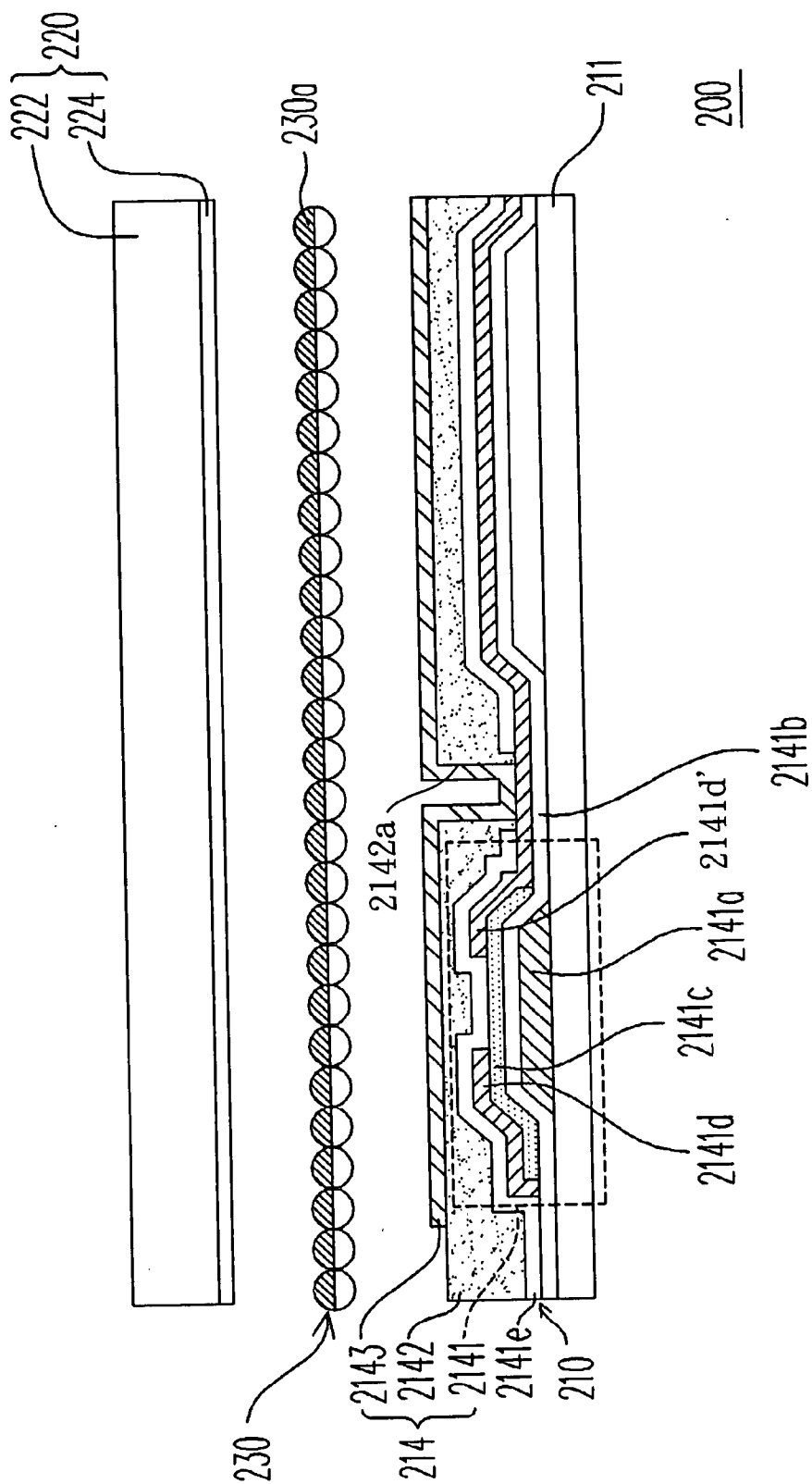


Fig. 1B

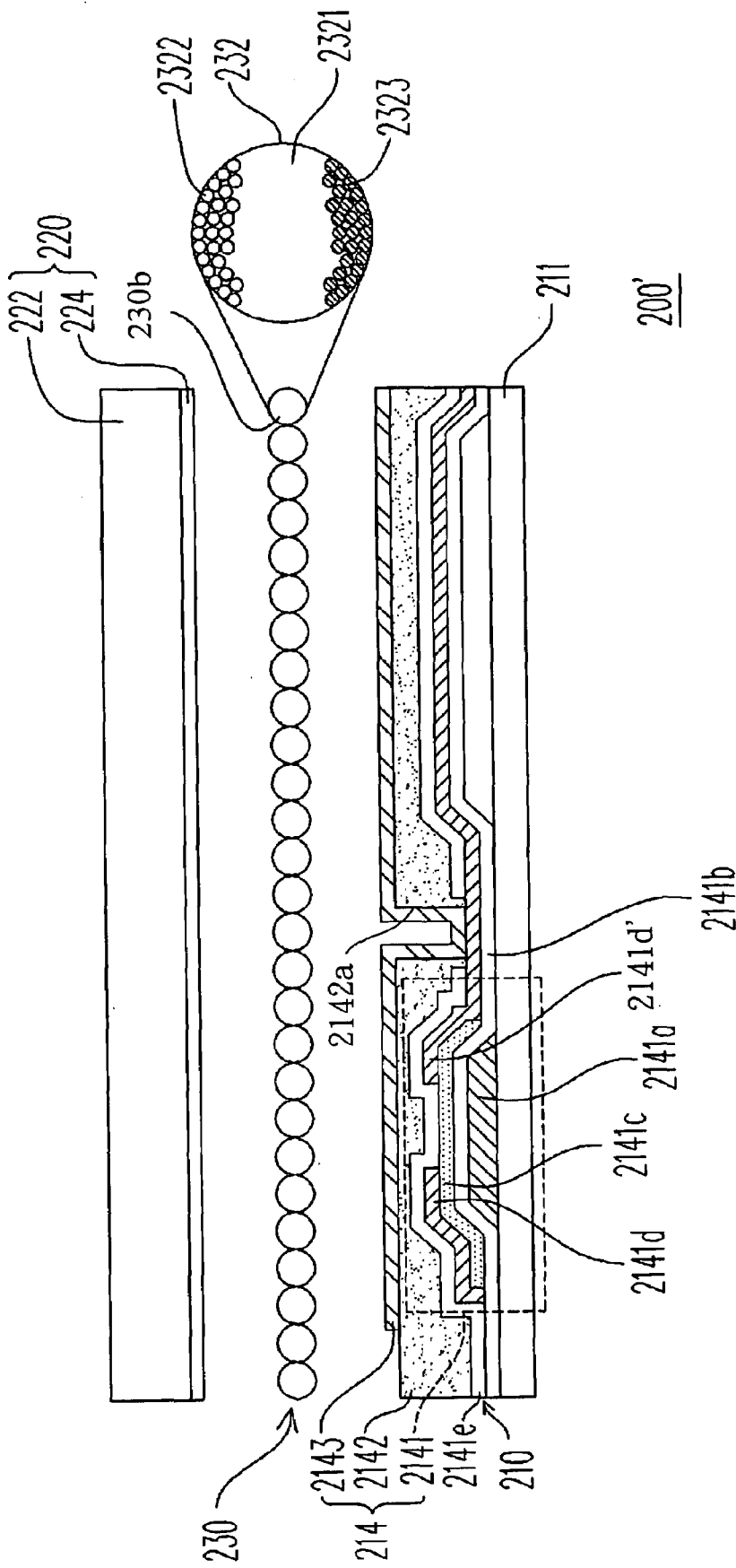


Fig. 1C

E-INK DISPLAY PANEL**RELATED APPLICATIONS**

[0001] The present application is based on, and claims priority from, Taiwan Application Serial Number 95130727, filed Aug. 21, 2006, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] 1. Field of Invention

[0003] The present invention relates to a display panel. More particularly, the present invention relates to an E-ink display panel.

[0004] 2. Description of Related Art

[0005] The E-ink display device of the first generation was developed in 1970. The first generation E-ink display device comprises many charged little balls. One side of the balls is white, and the other side thereof is black. When the applied electric field is changed, the balls will rotate to show white or black color. The E-ink display device of the second generation was developed in 1990. The second generation E-ink display device uses microcapsules to replace the charged balls stated above. The capsules filled with colored oil and white-colored particles inside. An external electrical field is applied to the white-colored particles to control their motions. When the white-colored particles move upward (in the direction toward the reader), the E-ink display device will show white color. When the white-colored particles move downward (in the direction away from the reader), the E-ink display device will show the color of the oil.

[0006] E-ink paper has as high contrast as regular paper and is easy to read. Moreover, E-ink paper has low power consumption, flexible, lightweight and portable. Because of these advantages, E-ink like a PDA, a mobile phone and an electronic reader has become a solution to be a highly readable and information intensive portable means in any dynamic light environment.

SUMMARY

[0007] An E-ink display panel is thus provided. In the E-ink display panel, the thickness of the dielectric layer which is disposed between the bottom-gate thin film transistors (TFTs) and the pixel electrodes prevents the pixel electrodes from being interfered by noise generated by the bottom-gate TFTs, the scan lines and the data lines. Therefore, the display quality of the E-ink display panel can be improved.

[0008] In accordance with the foregoing and other objectives of the present invention, an E-ink display panel is provided. The E-ink display panel comprises an active array substrate, an opposite substrate and an E-ink display medium. The active array substrate comprises a substrate, scan lines, data lines and pixel structures. The pixel structures are separately electrically connected with and driven by the scan lines and the data lines on the substrate. Each pixel structure comprises a bottom-gate TFT, a dielectric layer and a pixel electrode. The bottom-gate TFT comprises a gate electrode, a source electrode and a drain electrode. The gate electrode is electrically connected with one of the scan lines, and the source electrode is electrically connected with one of the data lines. The dielectric layer is disposed on the bottom-gate TFT. The dielectric layer has a contact window to expose a portion of the drain electrode of the

bottom-gate TFT, and the thickness of the dielectric layer is about 1~4 μm . The pixel electrode on the dielectric layer is electrically connected with the drain electrode through the contact window. The opposite substrate is disposed corresponding to the active array substrate. The E-ink display medium is disposed between the active array substrate and the opposite substrate.

[0009] In an embodiment, the pixel electrode covers the corresponding bottom-gate TFT and the data line that is electrically connected with the pixel electrode. In another embodiment, the pixel electrode further covers the scan line that is electrically connected with the pixel electrode. In still another embodiment, the pixel electrode covers the adjacent scan lines and/or data lines that are not electrically connected with the pixel electrode.

[0010] In conclusion, the dielectric layer with a thickness of about 1~4 μm is disposed between the bottom-gate TFTs of the active array substrate and the pixel electrodes in the invention. The thickness of the dielectric layer can prevent the pixel electrodes from being interfered by noise generated by the bottom-gate TFTs, the scan lines and the data lines. Therefore, the display quality of the E-ink display panel can be improved. Moreover, the thickness of the dielectric layer can make the surface of the active array substrate even flatter.

[0011] These and other features, aspects, and advantages of the present invention will become better understood with reference to the following description and appended figures.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention. In the drawings,

[0013] FIG. 1A is a top view of an active array substrate of an E-ink display panel according to an embodiment of the invention;

[0014] FIG. 1B is a cross-section view of an active array substrate of an E-ink display panel of FIG. 1A along line A-A'; and

[0015] FIG. 1C is a structural diagram of an E-ink display panel according to another embodiment of the invention.

DETAILED DESCRIPTION

[0016] FIG. 1A is a top view of an active array substrate of an E-ink display panel according to an embodiment of the invention. FIG. 1B is a cross-section view of an active array substrate of an E-ink display panel of FIG. 1A along line A-A'. FIG. 1A and FIG. 1B show only one pixel of the E-ink display panel. Please refer to FIG. 1B, the E-ink display panel 200 comprises mainly an active array substrate 210, an opposite substrate 220 and a display medium 230. Structure of elements and correlation thereof will be illustrated below with figures.

[0017] Please refer to FIG. 1A and FIG. 1B. The active array substrate 210 comprises a substrate 211, scan lines 212, data lines 213 and pixel structures 214. The substrate 211 can be a glass substrate, a plastic substrate or other types of substrates. The scan lines 212 and the data lines 213 are perpendicular to each other on the substrate 211. The pixel structures 214 are arranged in a matrix to define pixel

regions P. Moreover, the pixel structures **214** are electrically connected with and driven by the scan lines **212** and the data lines **213**. Each pixel structure **214** comprises a bottom-gate thin film transistor (TFT) **2141**, a dielectric layer **2142** and a pixel electrode **2143**.

[0018] A bottom-gate TFT **2141** comprises a gate electrode **2141a**, a gate isolation layer **2141b**, a channel layer **2141c**, a source electrode **2141d** and a drain electrode **2141d'**. The gate electrode **2141a** is disposed on the substrate **211** and is electrically connected with the scan line **212**. The gate isolation layer **2141b** is disposed on the substrate **211** and covers the gate electrode **2141a**. The channel layer **2141c** is disposed on the gate isolation layer **2141b** above the gate electrode **2141a**. The source electrode **2141d** and the drain electrode **2141d'** are at two sides of the channel layer **2141c**. The drain electrode **2141d'** is electrically connected with the data line **214**. A protection layer **2141e** can be selectively formed on the gate-bottom TFT **2141**, the scan line **212** and the data line **231** to protect underlying elements from damage and moisture. The material of the protection layer **2141e** is usually silicon nitride (Si_3N_4) or silicon oxide. The thickness of the protection layer **2141e** is a few thousand angstroms.

[0019] The dielectric layer **2142** with a thickness of about 1~4 μm is disposed on the gate-bottom TFT **2141**. The dielectric layer **2142** covers the whole substrate **211** including the data lines **212** and the scan lines **213**. The thickness of the dielectric layer **2142** is capable to shield the noise interference from bottom-gate TFTs **2141** underlying the pixel electrodes **2143**, the scan lines **212** and the data lines **213**. Therefore, noise influence on the pixel electrode **2143** can be ignored. The dielectric layer **2142** is made from a photo resist material or a resin material. When the dielectric layer **2142** is made from a photo resist material, the dielectric layer **2142** can be defined to form contact windows **2142a** by photolithography.

[0020] The contact window **2142a** in the dielectric layer **2142** exposes a portion of the drain electrode **2142d'** of the bottom-gate TFT **2141**. The pixel electrode **2143** on the dielectric layer **2142** is electrically connected with the drain electrode **2141d'** of the bottom-gate TFT **2141** through the contact window is **2142a** in the dielectric layer **2142**. The material of the pixel electrode **2143** is indium tin oxide (ITO) or indium zinc oxide (IZO). In the embodiment, the pixel electrode **2143** completely covers the corresponding bottom-gate TFT **2141** and the data line **213** that is electrically connected with the pixel electrode **2143**. In another embodiment, the pixel electrode **2143** can cover the scan line **212** that is electrically connected with the pixel electrode **2143**. In still another embodiment, the pixel electrode **2143** can further covers the scan lines and/or the data lines that are not electrically connected with the pixel electrode **2143**.

[0021] In FIG. 1B, the opposite substrate **220** is disposed correspondingly to the active array substrate **210**. The opposite substrate **220** comprises a substrate **222** and a common electrode **224**. The common electrode **224** can be a transparent conductive layer. The display medium **230** is disposed between the active array substrate **210** and the opposite substrate **220**.

[0022] The display medium **230** is at least bi-stable. The display medium **230** can keep display images even the applied signal is removed. In the embodiment, the display medium comprises many E-ink particles **230a**. One side of

each E-ink particle **230a** is light-colored, and the other side thereof is dark-colored. The two separate sides of the E-ink particles **230a** have different electric characteristics. When the electric field between the pixel electrode **2143** and the common electrode **224** is changed, the E-ink particles **230a** will be driven to display images on the E-ink display panel **200**.

[0023] The display medium **230** is not limited to the type stated above. FIG. 1C is a structural diagram of an E-ink display panel according to another embodiment of the invention. In the E-ink display panel **200'** of FIG. 1C, display medium **230** comprises dark-colored particles **2323**, light-colored particles **2322** and a transparent fluid **2321**. The dark-colored particles **2323** and the light-colored particles **2322** are distributed in the transparent fluid **2321** and have different electrical characteristics. When the electrical field between the pixel electrode **2143** and the common electrode **224** is changed, the dark-colored particles **2323** and the light-colored particles **2322** will move upward and downward according to the direction of the electrical field and thus to enable each pixel to show the colors of different particles. In still another embodiment, the display medium **230** comprises single charge colored particles suspended in a colored liquid. The single charge colored particles will move upward or downward according to the direction of the electric field and thus to enable each pixel to show the colors of the colored particles and the colored liquid.

[0024] In a further embodiment, the display medium **230** is wrapped inside microcapsules. In another embodiment, the display medium **230** is contained in microcups. In an embodiment, the display medium **230** can move in the active regions freely and is not limited to a side structural body. In other embodiments, the display medium **230** can be arranged according to different structures. The type and arrangement of the display medium **230** is not limited by the above embodiments.

[0025] In conclusion, the dielectric layer **2142** with a thickness of about 1~4 μm is disposed between the bottom-gate TFTs of the active array substrate and the pixel electrodes in the E-ink display panel according to the embodiments of the invention. The dielectric layer can shield the noise interference from the bottom-gate TFTs, the scan lines and the data lines. Therefore, the noise interference on the pixel electrodes is greatly reduced, and the display quality of the E-ink display panel can be improved. Moreover, the thickness of the dielectric layer can make the surface of the active array substrate even flatter.

[0026] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims and their equivalents.

What is claimed is:

1. An E-ink display panel, comprising:
an active array substrate, comprising:
a substrate;
a plurality of scan lines and a plurality of data lines disposed on the substrate;
a plurality of pixel structures electrically connected with and driven by the scan lines and the data lines respectively, wherein each of the pixel structures, comprising:

- a bottom-gate thin film transistor (TFT), which comprises a gate electrode, a source electrode and a drain electrode, wherein the gate electrode is electrically connected with one of the scan lines, and the source electrode is electrically connected with one of the data lines;
 - a dielectric layer disposed on the bottom-gate TFT, wherein the dielectric layer has a contact window to expose a portion of the drain electrode, and the thickness of the dielectric layer is about 1~4 μm ; and
 - a pixel electrode disposed on the dielectric layer and electrically connected with the drain electrode through the contact window;
- an opposite substrate disposed corresponding to the active array substrate; and
- an E-ink display medium disposed between the active array substrate and the opposite substrate.
2. The E-ink display panel of claim 1, wherein the active array substrate further comprises a protection layer to cover the bottom-gate TFTs, the scan lines and the data lines.
3. The E-ink display panel of claim 2, wherein the material of the protection layer is silicon nitride or silicon oxide.
4. The E-ink display panel of claim 1, wherein the dielectric layer is made from a photo resist material or a resin material.
5. The E-ink display panel of claim 1, wherein the dielectric layer covers the scan lines and the data lines.
6. The E-ink display panel of claim 5, wherein the pixel electrode covers the data line that is electrically connected with the pixel electrode.
7. The E-ink display panel of claim 5, wherein the pixel electrode covers the scan line that is electrically connected with the pixel electrode.
8. The E-ink display panel of claim 5, wherein the pixel electrode covers the adjacent data lines that are not electrically connected with the pixel electrode.

9. The E-ink display panel of claim 5, wherein the pixel electrode covers the adjacent scan lines that are not electrically connected with the pixel electrode.

10. The E-ink display panel of claim 1, wherein the pixel electrode completely covers the corresponding bottom-gate TFT.

11. The E-ink display panel of claim 1, wherein the opposite substrate, comprising:

- a substrate; and
- a common electrode disposed between the substrate and the E-ink display medium.

12. The E-ink display panel of claim 1, wherein the E-ink display medium, comprising:

- a plurality of dark-colored particles;
- a plurality of light-colored particles; and
- a transparent fluid, wherein the dark-colored particles and the light-colored particles are distributed in the transparent fluid and have different electric characteristics.

13. The E-ink display panel of claim 1, wherein the E-ink display medium, comprising a plurality of E-ink particles, wherein one side of the E-ink particles is light-colored, the other side thereof is dark-colored, and those two sides have different electric characteristics.

14. The E-ink display panel of claim 1, wherein the E-ink display medium, comprising:

- a plurality of colored particles; and
- a colored liquid, wherein the colored particles are suspended in the colored liquid.

15. The E-ink display panel of claim 1, wherein the E-ink display medium is wrapped inside a plurality of microcapsules.

16. The E-ink display panel of claim 1, wherein the E-ink display medium is contained in a plurality of microcaps.

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